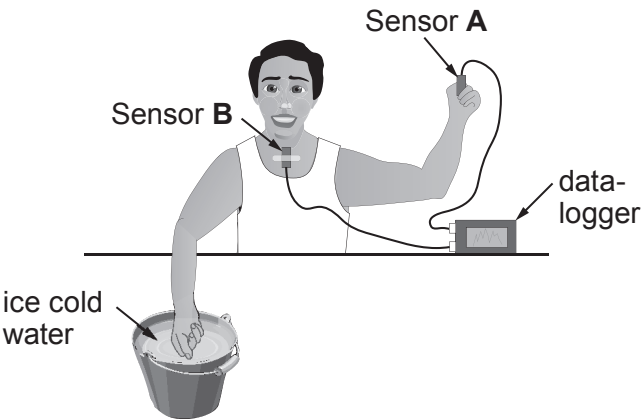


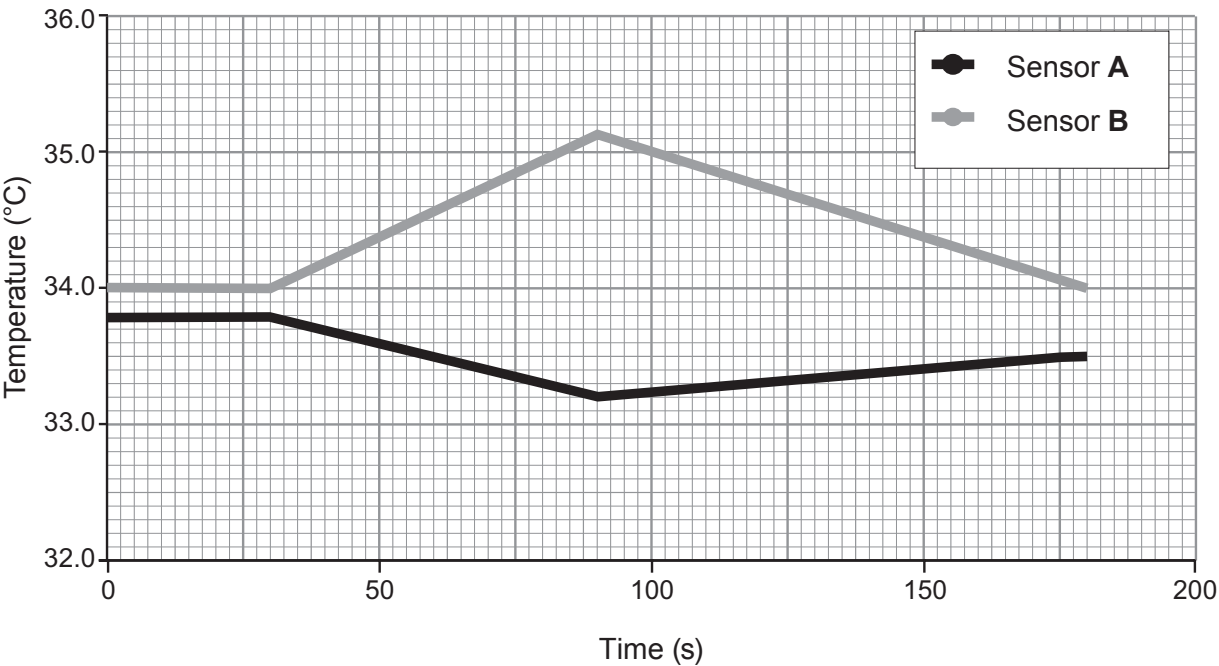
9. Dylan set up an experiment to study negative feedback mechanisms associated with rapid cooling of the hand.

A temperature sensor was held between the fingertips of a volunteer and another sensor was taped to the skin on the upper chest. The sensors were connected to a data logger that was set to record for three minutes.



The volunteer placed their right hand into ice cold water 30 seconds after the recording started. The hand was then removed from the ice cold water after a further 30 seconds.

The results from the data logger are shown below:



Examiner
only

(a) State what is meant by a negative feedback mechanism. [2]

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(b) (i) Describe the results for sensor **A** after the hand was placed in the ice cold water. [1]

(ii) Explain the change in temperature recorded by sensor **A** between 30 seconds and 90 seconds. [4]

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(c) Suggest the reason for the increase in the temperature recorded by sensor **B** between 30 seconds and 90 seconds. [1]

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(d) Suggest how the results of the experiment would differ if the volunteer had consumed an alcoholic drink 20 minutes before the start of the experiment. [1]

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9

